

Laboratory work 3

We continue to work with the database from the previous laboratory works.

Take a full-page screenshot that covers the code and results of each task.

DML

1. Generate and insert 200 random rows in your airport database.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure with 'lab2' selected. The central query editor contains the following SQL code:

```
1 INSERT INTO airline1
2 (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)
3 SELECT
4 1000 + i,
5 'A' || i,
6 'Random_airline' || i,
7 'Country' || i,
8 CURRENT_TIMESTAMP,
9 CURRENT_TIMESTAMP
10 FROM generate_series(1, 200) as i;
11
12 SELECT * FROM airline1;
13
```

The bottom pane shows the results of the query, displaying a table with 7 columns: airline_id, airline_code, airline_name, airline_country, created_at, and updated_at. The table contains 205 rows, with the first few rows visible:

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
1012	A12	Random_airline12	Country12	2026-09-28 12:02:23.630343	2026-09-28 12:02:23.630343
1013	A13	Random_airline13	Country13	2026-09-28 12:02:23.630343	2026-09-28 12:02:23.630343
1014	A14	Random_airline14	Country14	2026-09-28 12:02:23.630343	2026-09-28 12:02:23.630343
1015	A15	Random_airline15	Country15	2026-09-28 12:02:23.630343	2026-09-28 12:02:23.630343
1016	A16	Random_airline16	Country16	2026-09-28 12:02:23.630343	2026-09-28 12:02:23.630343
1017	A17	Random_airline17	Country17	2026-09-28 12:02:23.630343	2026-09-28 12:02:23.630343

Total rows: 205 Query complete 00:00:00.164

2. Add a new airline named "KazAir" based in "Kazakhstan" to the airline table.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure with 'lab2' selected. The central query editor contains the following SQL code:

```
1 INSERT INTO airline1
2 (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)
3 VALUES
4 (3000, 'KZ', 'KazAir', 'Kazakhstan', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
5
6 SELECT * FROM airline1
7 WHERE airline_name = 'KazAir';
```

The bottom pane shows the results of the query, displaying a table with 7 columns: airline_id, airline_code, airline_name, airline_country, created_at, and updated_at. The table contains 1 row:

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
3000	KZ	KazAir	Kazakhstan	2026-09-28 12:12:16.443267	2026-09-28 12:12:16.443267

Total rows: 1 Query complete 00:00:00.216

3. Update the airline country "KazAir" to "Turkey".

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left shows the database structure. The Query editor in the center contains the following SQL code:

```
1 UPDATE airline1
2 SET airline_country = 'Turkey'
3 WHERE airline_name = 'KazAir';
4
5 SELECT * FROM airline1
6 WHERE airline_name = 'KazAir';
```

The Data Output pane at the bottom shows the results of the query. It displays a table with 7 columns: airline_id, airline_code, airline_name, airline_country, created_at, and updated_at. The results show one row for 'KazAir' with the country updated to 'Turkey'.

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
3000	KZ	KazAir	Turkey	2026-09-28 12:12:16.443267	2026-09-28 12:12:16.443267

Successfully run. Total query runtime: 190 msec. 1 rows affected.

4. Add three airlines at once: "AirEasy" in "France", "FlyHigh" in "Brazil" and "FlyFly" in "Poland".

The screenshot shows the pgAdmin 4 interface. The Query editor in the center contains the following SQL code:

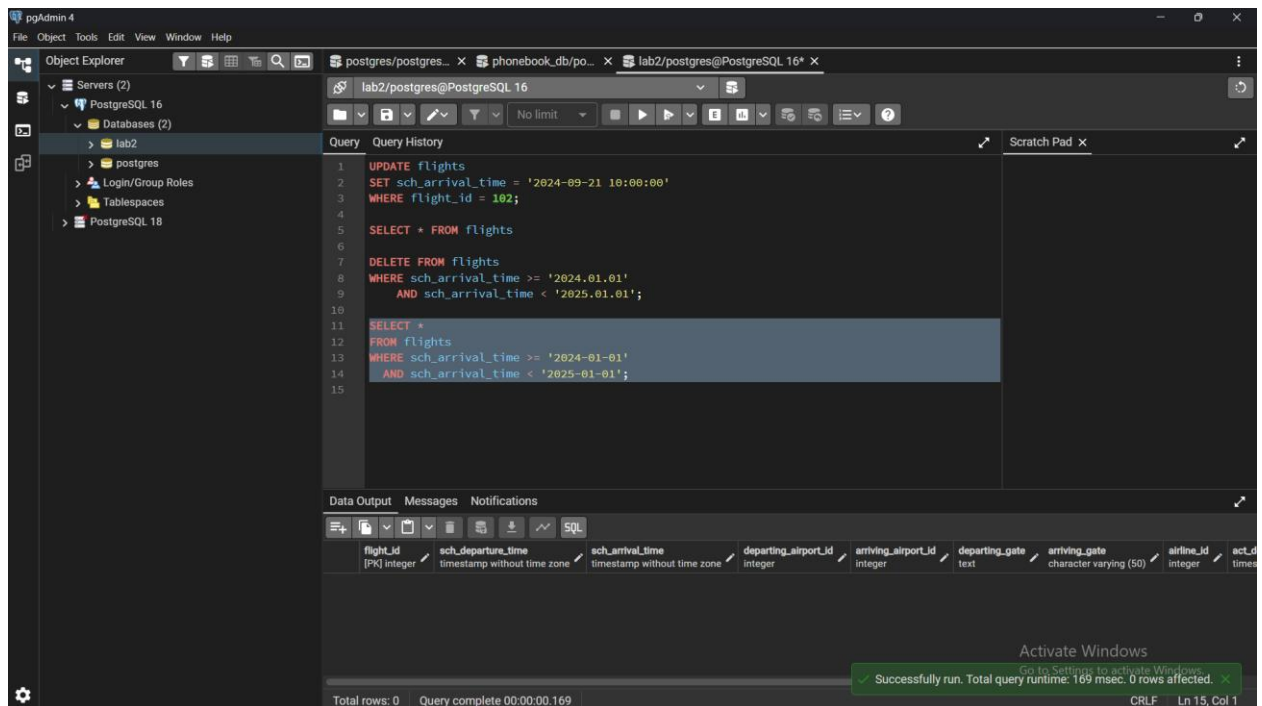
```
1 INSERT INTO airline1
2 (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)
3 VALUES
4 (3001, 'AE', 'AirEasy', 'France', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
5 (3002, 'FH', 'FlyHigh', 'Brazil', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
6 (3003, 'FF', 'FlyFly', 'Poland', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
7
8 SELECT *
9 FROM airline1
10 WHERE airline_name IN ('AirEasy', 'FlyHigh', 'FlyFly');
```

The Data Output pane at the bottom shows the results of the query. It displays a table with 7 columns: airline_id, airline_code, airline_name, airline_country, created_at, and updated_at. The results show three rows for 'AirEasy', 'FlyHigh', and 'FlyFly'.

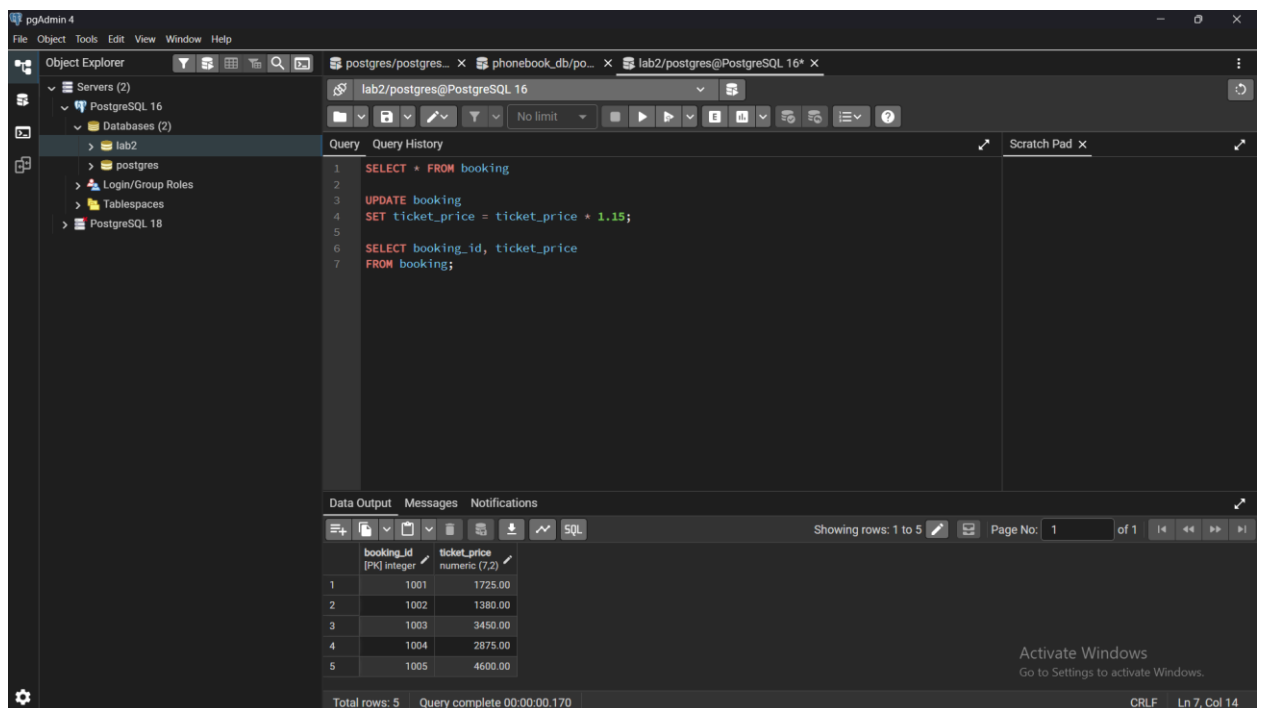
airline_id	airline_code	airline_name	airline_country	created_at	updated_at
3001	AE	AirEasy	France	2026-09-28 12:17:18.045035	2026-09-28 12:17:18.045035
3002	FH	FlyHigh	Brazil	2026-09-28 12:17:18.045035	2026-09-28 12:17:18.045035
3003	FF	FlyFly	Poland	2026-09-28 12:17:18.045035	2026-09-28 12:17:18.045035

Successfully run. Total query runtime: 186 msec. 3 rows affected.

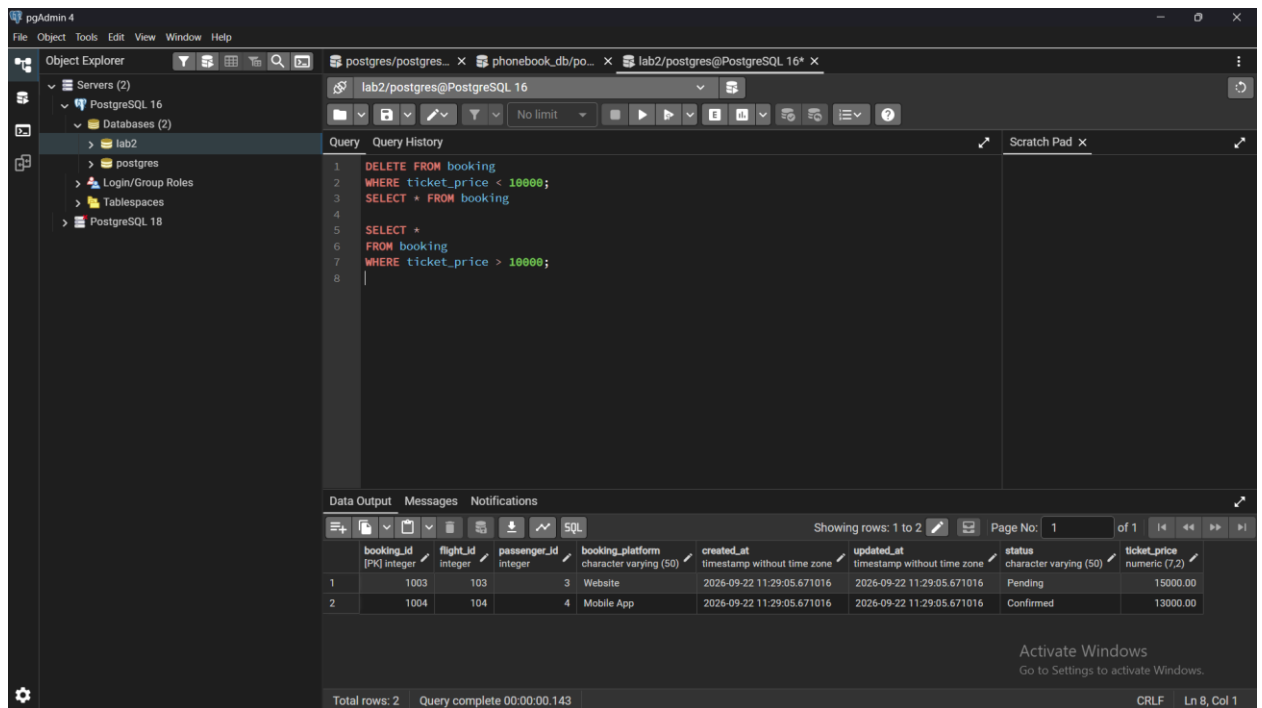
5. Delete all flights whose arrival in 2024 year.



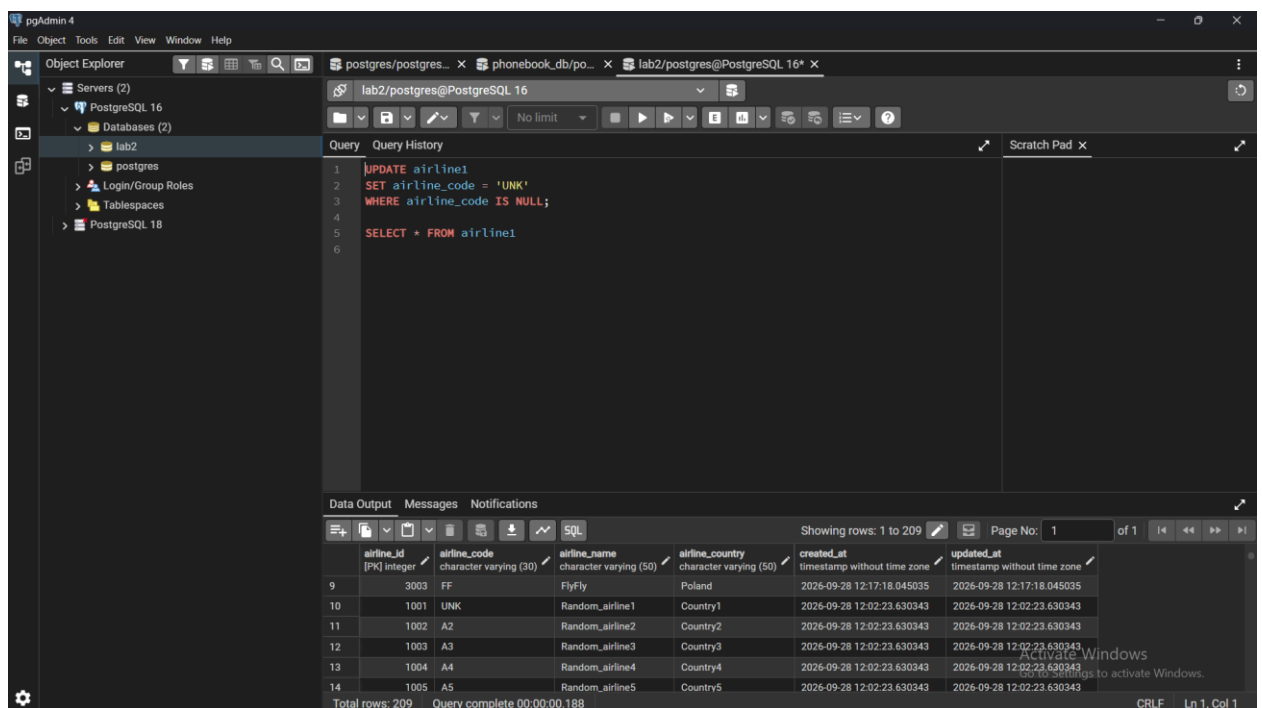
6. Increase the price of all tickets in booking table for flights by 15%.



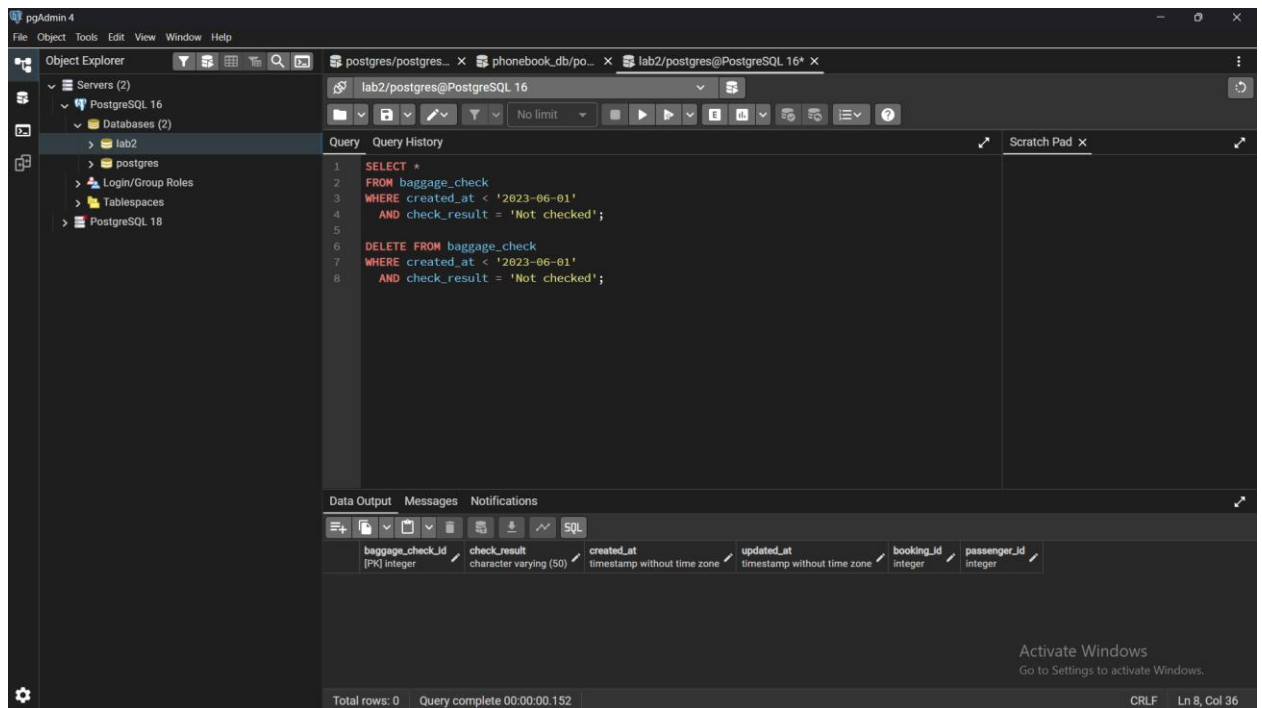
7. Delete all tickets where price is less than 10000.



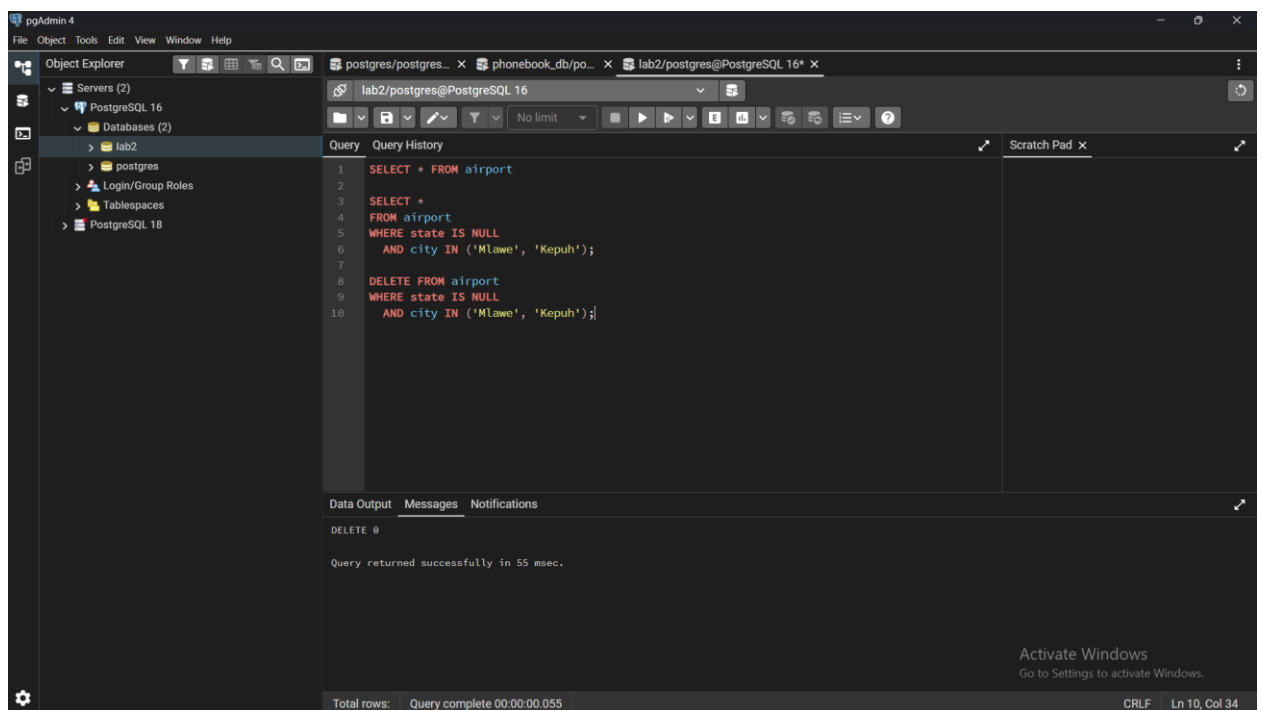
8. Set a default code 'UNK' (Unknown) for any airline in the airline table where airline_code is missing (NULL).



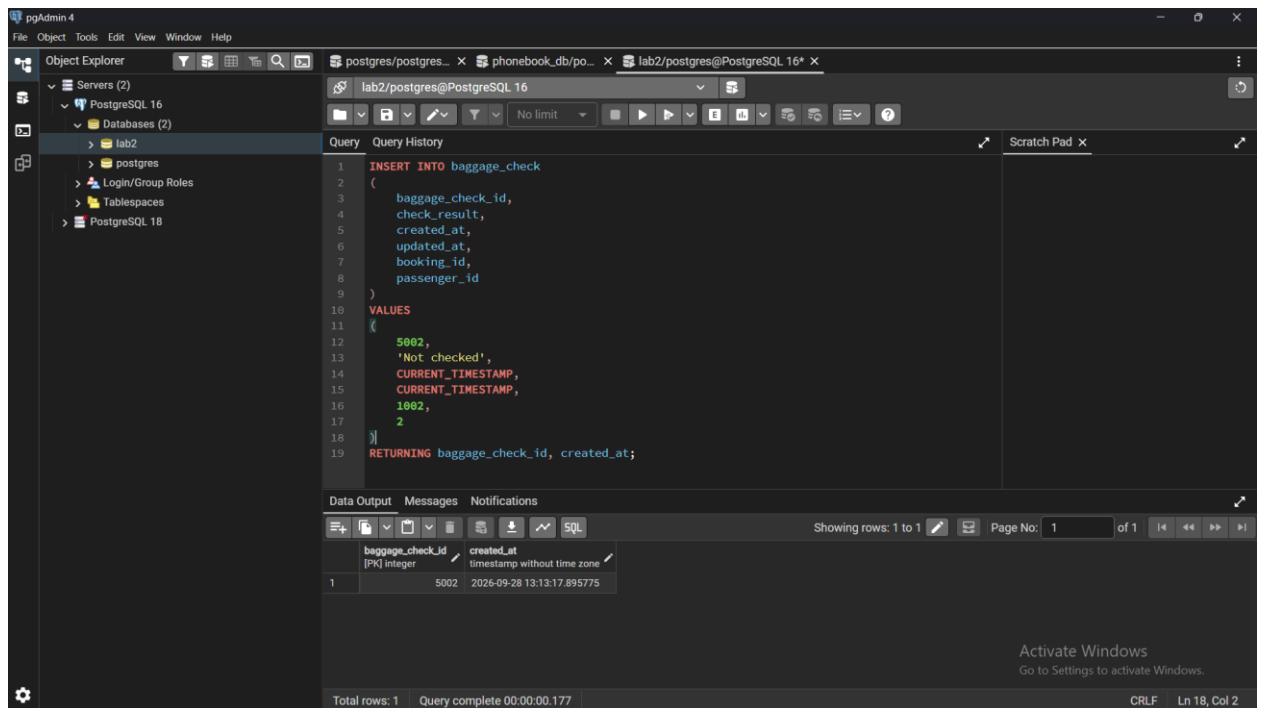
9. Delete all baggage check records created before June 1, 2023, whose result is still 'Not checked'.



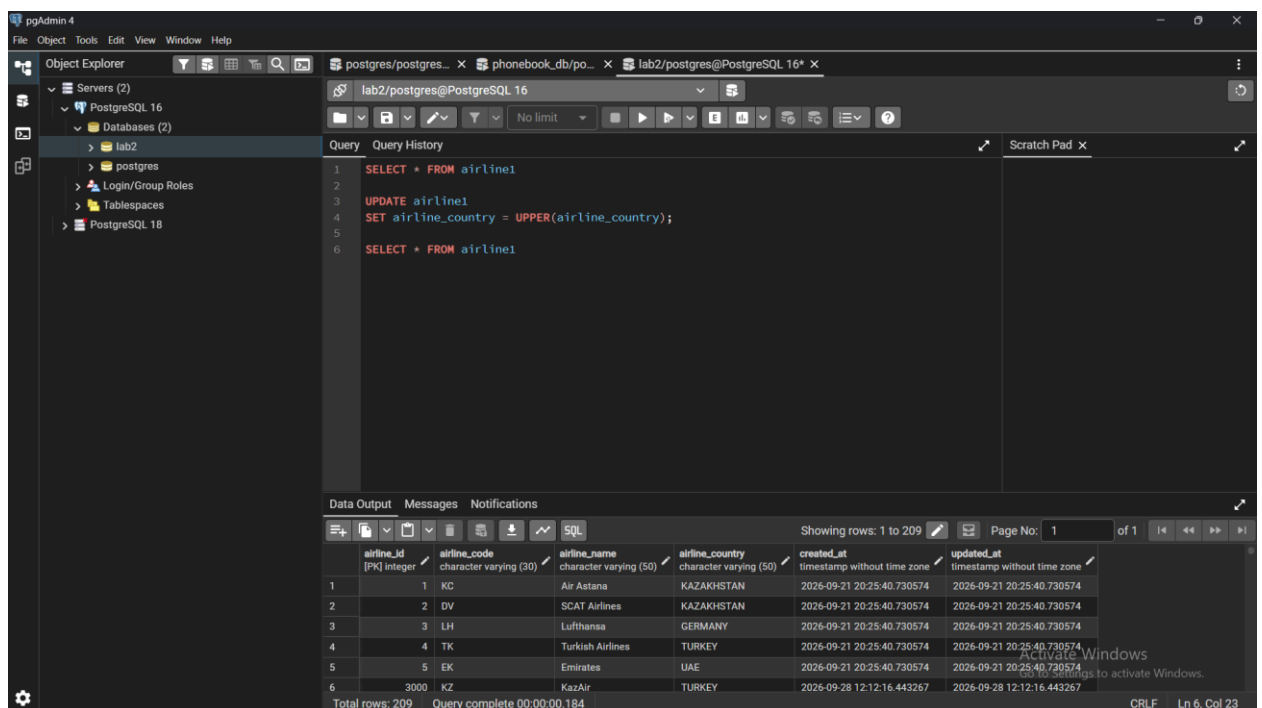
10. Delete all entries from the airport table where state is missing (NULL) and the city is either 'Mlawe' or 'Kepuh'.



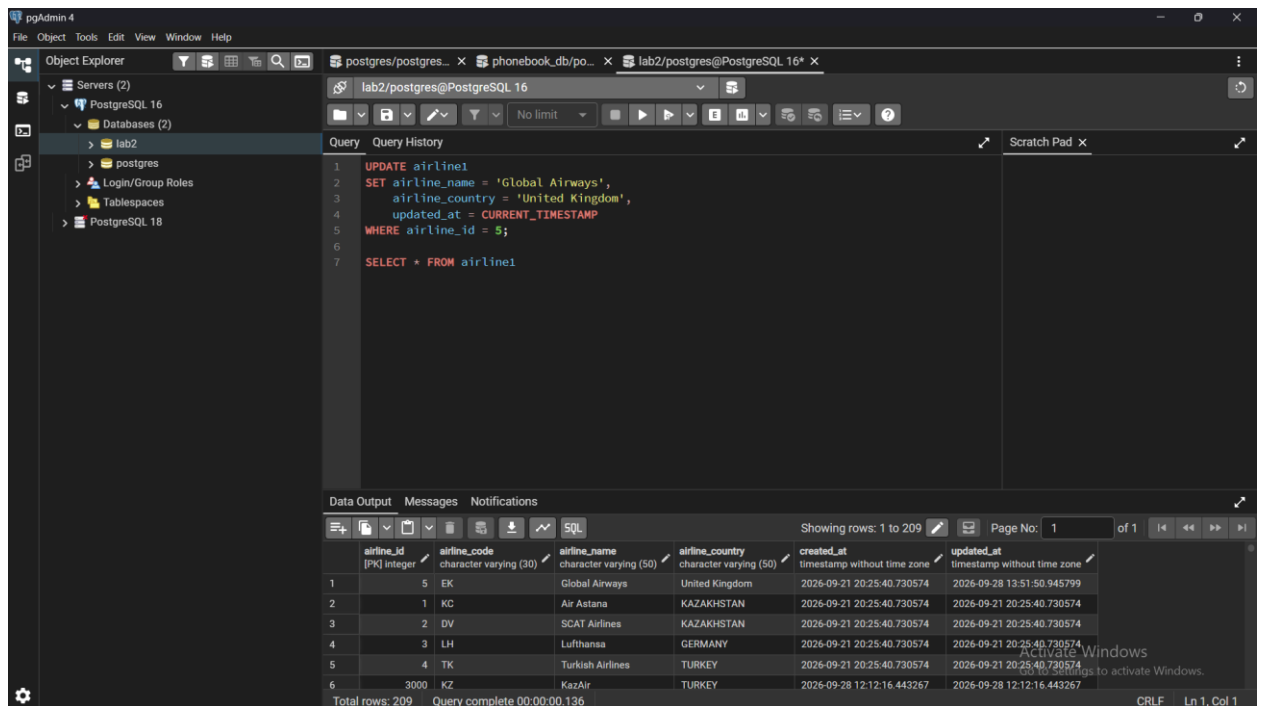
11. Insert a new unchecked baggage inspection record and immediately return its newly generated baggage_check_id and created_at timestamp.



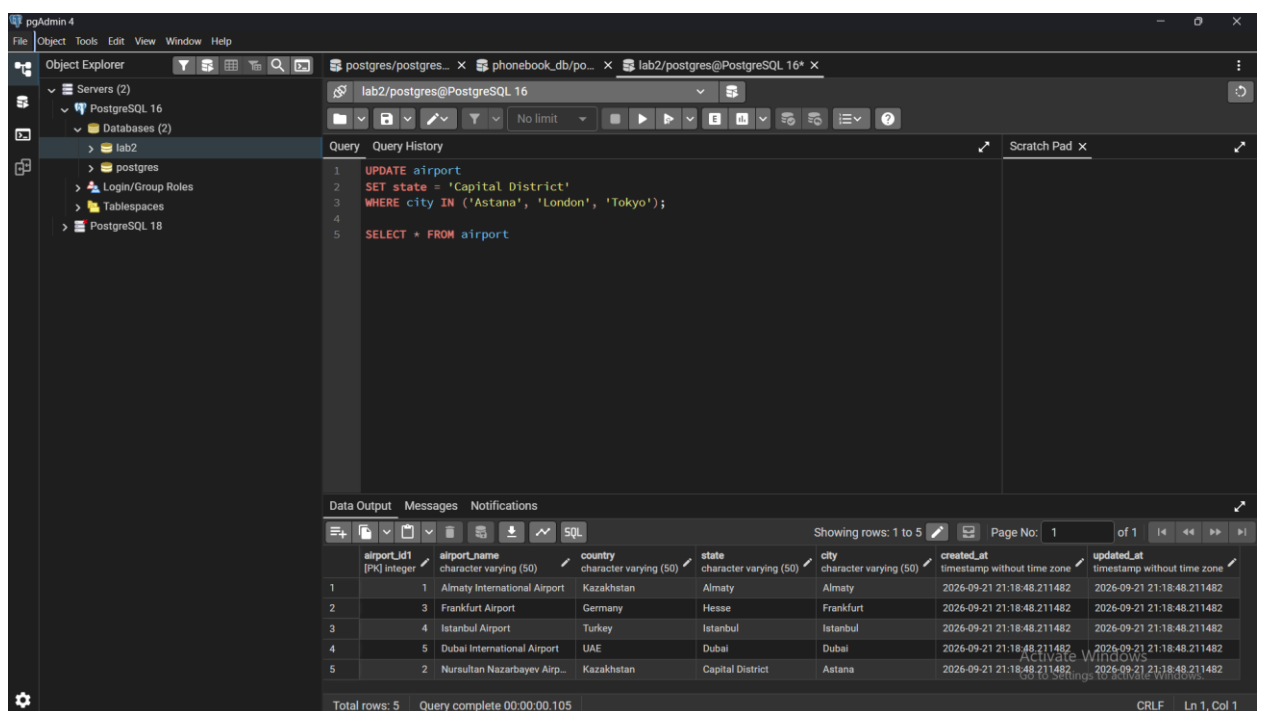
12. Convert all country names in the airline table to uppercase to ensure data consistency.



13. Update both the name and country for the airline with airline_id = 5. Set the name to 'Global Airways' and the country to 'United Kingdom', while updating update_at to the current date.



14. Set the state field to 'Capital District' for airports located in specific cities: 'Astana', 'London', and 'Tokyo'



15. Mark all baggage check records (baggage_check) created in March 2024 with a status of 'Not checked' as completed ('Checked').

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Servers (2)
 - PostgreSQL 16
 - Databases (2)
 - lab2
 - postgres
 - Login/Group Roles
 - Tablespaces
 - PostgreSQL 18

lab2/postgres@PostgreSQL 16

Query

```
1 UPDATE baggage_check
2 SET check_result = 'Checked'
3 WHERE created_at >= '2024-03-01'
4 AND created_at < '2024-04-01'
5 AND check_result = 'Not checked';
6
7 SELECT * FROM baggage_check
```

Scratch Pad

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

baggage_check_id [PK] integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	booking_id integer	passenger_id integer
1	3 Checked	2023-05-15 14:00:00	2023-05-15 14:00:00	1003	3
2	4 Not checked	2024-01-10 09:00:00	2024-01-10 09:00:00	1004	4
3	5001 Not checked	2026-09-28 13:10:53.638353	2026-09-28 13:10:53.638353	1001	1
4	5002 Not checked	2026-09-28 13:13:17.895775	2026-09-28 13:13:17.895775	1002	2

Total rows: 4 Query complete 00:00:00.103 CRLF Ln 2, Col 8

Activate Windows
Go to Settings to activate Windows.